Designation: A 3456-24

Standard Specification for High-Strength Low-Alloy Structural Steel Plate

This standard was issued on January 30, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers high-strength low-alloy structural steel plates for construction and heavy equipment applications

1.2 The HSLA structural steel plates covered by this specification are intended for use in bridges, buildings, pressure vessels, mining equipment, and other structural applications requiring high strength and good weldability.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

A 6 Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling

A 370 Test Methods and Definitions for Mechanical Testing of Steel Products

E 23 Test Methods for Notched Bar Impact Testing of Metallic Materials

E 3 Guide for Preparation of Metallographic Specimens

3. Terminology

3.1 Definitions:

3.1.1 high-strength low-alloy steel—steel with enhanced mechanical properties achieved through microalloying and controlled processing

3.1.2 normalized—heat treatment involving heating above the upper critical temperature followed by air cooling

3.1.3 Charpy V-notch energy—measure of material toughness determined by impact testing of notched specimens

4. Classification

4.1 HSLA structural steel plates shall be classified according to minimum yield strength and thickness range.

4.1.1 **Grade 50**—Minimum yield strength 345 MPa, thickness 6-50 mm

4.1.2 **Grade 60**—Minimum yield strength 415 MPa, thickness 6-40 mm

4.1.3 **Grade 70**—Minimum yield strength 485 MPa, thickness 6-32 mm

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Grade (50, 60, or 70)

5.1.3 Dimensions (length × width × thickness in mm)

5.1.4 Quantity (number of plates or total weight in tonnes)

5.1.5 Heat treatment condition

5.1.6 Impact testing temperature (if required)

6. Materials and Manufacture

6.1 Steel shall be killed and produced to fine grain practice.

6.2 Material may be supplied in hot-rolled, normalized, or thermomechanically processed condition.

6.3 Chemical composition shall be optimized for weldability and shall meet requirements in Section 7.

7. Chemical Composition

7.1 The material shall conform to the chemical composition requirements specified in Table 1.

Element	Composition, %
Carbon (C), max	0.18%
Manganese (Mn)	0.80-1.50%
Phosphorus (P), max	0.025%
Sulfur (S), max	0.015%
Silicon (Si)	0.15-0.50%
Niobium (Nb)	0.02-0.05%
Vanadium (V)	0.03-0.08%
Copper (Cu), max	0.35%

8. Mechanical and Physical Requirements

Property	Requirement	Test Method
Yield strength (Grade 50)	≥ 345 MPa	ASTM A 370
Tensile strength	450-620 MPa	ASTM A 370

Elongation (50 mm)	≥ 18%	ASTM A 370
Charpy V-notch (-20°C)	≥ 27 J avg	ASTM E 23

9. Dimensions and Permissible Variations

9.1 Thickness tolerance shall conform to ASTM A 6 for plates.

9.2 Width and length tolerances shall be per ASTM A 6 unless otherwise specified.

9.3 Flatness shall meet requirements of ASTM A 6 for the applicable thickness.

10. Workmanship, Finish, and Appearance

10.1 Plates shall be free from injurious defects. Surface imperfections may be removed by grinding provided minimum thickness is maintained.

11. Sampling

11.1 One set of mechanical tests shall be performed per heat and thickness range. Impact tests required when specified.

12. Test Methods

12.1 **Tensile Testing**—Machine longitudinal tensile specimens per ASTM A 370 and test to determine yield strength, tensile strength, and elongation.

12.2 **Impact Testing**—Prepare Charpy V-notch specimens per ASTM E 23 and test at specified temperature.

12.3 **Chemical Analysis**—Perform ladle or product analysis to verify chemistry compliance.

13. Inspection

13.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

14. Rejection and Rehearing

14.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

15. Certification

15.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

16. Product Marking

16.1 Plates shall be stenciled or tagged with heat number, grade, thickness, and manufacturer identification.

17. Packaging and Package Marking

17.1 Plates shall be bundled and strapped for shipment. Coating or wrapping may be specified to prevent corrosion during transit.

18. Keywords

18.1 steel; HSLA; structural; plate; high strength; weldable; construction